MILITARY SPECIFICATION SHEET

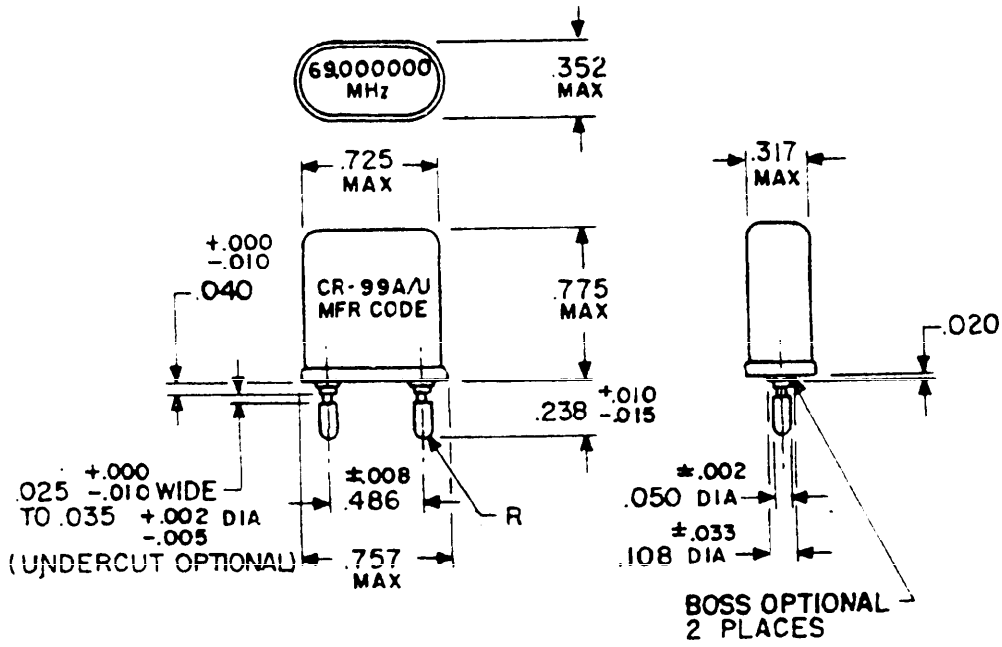
CRYSTAL UNIT, QUARTZ, CR-99A/U

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(D) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 44 to 69 MHz; third mechanical overtone; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.108	2.74	.486	12.34
.008	.20	.033	.84	.238	6.05	.725	18.42
.010	.25	.035	.89	.317	8.05	.757	19.23
.015	.38	.040	1.02			.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivaletns are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Crystal unit - CR-99A/U.

(D) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure 1
Frequency range	44 to 69 MHz, inclusive
Antiresonance, load capacitance	32.0 ±0.5pF
Mode of oscillation	Third mechanical overtone
Operating temperature range	-55°C to +105°C, inclusive
Frequency tolerance	±0.005 percent
Equivalent resistance	40 ohms, maximum
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4 mW
Calibration values	
Resistance	40 ohms
Voltage drop	0.28 volt
Capacitance, shunt	7 pF, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 ppm

1/ Method 201 of MIL-STD-202.

D QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTES:

1. This crystal-unit type is to be used for special applications, and is not for general use. Specific approval of the Government is required prior to each intended use.

2. These crystal units are for use in the BC-639 equipment. They were previously fabricated to the requirements specified in Air force Technical Order 12-1-20, Supplement 3A.

3. Formula for computing the crystal frequencies applicable to the Bendix BC-639 equipment is:

$$F_x = \frac{F_0 - 12\ \text{MHz}}{2}$$

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Ⓓ Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 85

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